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Task1

1)

```
module ct1(A,B,C,D,E,F,sel,out,out_bar);
input A,B,C,D,E,F,sel;
output out,out_bar;
wire w1,w2;

assign w1=A&B&C;
assign w2=~(D^E^F);
assign out=sel?w2:w1;
assign out_bar=~out;

endmodule
```

2)

```
module ct2(A,B,C,D,sel,out,out_bar);

input A,B,C,sel;
input [2:0]D;
output reg out,out_bar;
reg t1,t2;

always @(*)
begin
t1=D[0]&D[1];
t2=t1|D[2];

if(sel)
    out=~(A^B^C);
else
    out=t2;

out_bar=~out;
end

endmodule
```

3)

```
module adder(X,Y,Z);
input [3:0]X,Y;
output [3:0]Z;

assign Z=X+Y;

endmodule
```

4)

```
module decoder(in,out);
input [1:0]in;
output [3:0]out;

assign out=4'b1<<in;

endmodule
```

5)

```
module parity_gen(d,p);

input [7:0]d;
output [8:0]p;

assign p={d,^d};

endmodule
```

6)

```
module seg7(A,B,op,en,a,b,c,d,e,f,g);

input [3:0]A,B;
input [1:0]op;
input en;
output reg a,b,c,d,e,f,g;

wire [3:0]res;
reg [6:0]disp;

alu_4bit u1(.A(A),.B(B),.op(op),.R(res));

always @(*)
begin
if(~en)
    disp=7'd0;
else
    case(res)
    0:disp=7'b1111110;
    1:disp=7'b0110000;
    2:disp=7'b1101101;
    3:disp=7'b1111001;
    4:disp=7'b1010011;
    5:disp=7'b1011011;
    6:disp=7'b1011111;
    7:disp=7'b1110000;
    8:disp=7'b1111111;
    9:disp=7'b1111011;
    10:disp=7'b1110111;
    11:disp=7'b0011111;
    12:disp=7'b01101110;
    13:disp=7'b0111101;
    14:disp=7'b01011111;
    15:disp=7'b01001111;
    default:disp=7'b0;
    endcase
end

always @(*)
begin
{a,b,c,d,e,f,g}=disp;
end

endmodule
```